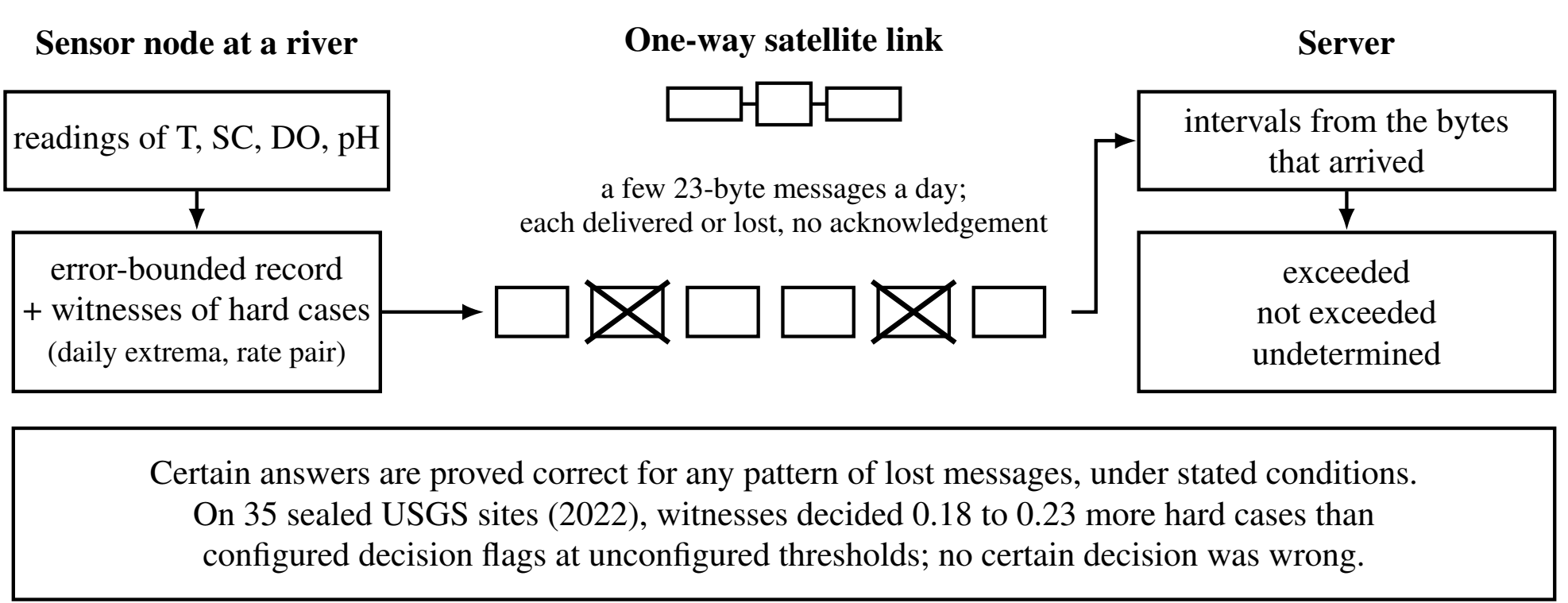
Sensor node at a river
One-way satellite link
Server
readings of T, SC, DO, pH
error-bounded record
+ witnesses of hard cases
(daily extrema, rate pair)
a few 23-byte messages a day;
each delivered or lost, no acknowledgement
intervals from the bytes
that arrived
exceeded
not exceeded
undetermined
Certain answers are proved correct for any pattern of lost messages, under stated conditions.
On 35 sealed USGS sites (2022), witnesses decided 0.18 to 0.23 more hard cases than
configured decision flags at unconfigured thresholds; no certain decision was wrong.